

Vector Algebra

$dl = (dx)\hat{x} + (dy)\hat{y} + (dz)\hat{z}$

$d\tau = dxdydz$

$\nabla t = \frac{\partial t}{\partial x}\hat{x} + \frac{\partial t}{\partial y}\hat{y} + \frac{\partial t}{\partial z}\hat{z}$

$\nabla \cdot v = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$

$\nabla \times v = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$

$\nabla^2 t = \frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2}$

$A \cdot (B \times C) = B \cdot (C \times A) = C \cdot (A \times B)$

$A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$

$\nabla(fg) = f(\nabla g) + g(\nabla f)$

$\nabla(A \cdot B) = A \times (\nabla \times B) + B \times (\nabla \times A) + (A \cdot \nabla)B + (B \cdot \nabla)A$

$\nabla \cdot (fA) = f(\nabla \cdot A) + A \cdot (\nabla f)$

$\nabla \cdot (A \times B) = B \cdot (\nabla \times A) - A \cdot (\nabla \times B)$

$\nabla \times (fA) = f(\nabla \times A) - A \times (\nabla f)$

$\nabla \times (A \times B) = (B \cdot \nabla)A - (A \cdot \nabla)B + A(\nabla \cdot B) - B(\nabla \cdot A)$

$\nabla \cdot (\nabla \times A) = 0$

$\nabla \times (\nabla f) = 0$

$\nabla \times (\nabla \times A) = \nabla(\nabla \cdot A) - \nabla^2 A$

Fundamental Theorems

$\int_a^b (\nabla f) \cdot dl = f(b) - f(a)$

$\int_V (\nabla \cdot A) d\tau = \int_S A \cdot da$

$\int_S (\nabla \times A) \cdot da = \int_P A \cdot dl$

Units

$1N = 1\left(\frac{\text{kg} \cdot m}{s^2}\right)$

$1J = 1(N \cdot m)$

$1W = 1\left(\frac{J}{s}\right)$

$1V = 1\left(\frac{J}{C}\right)$

$1F = 1\left(\frac{C}{V}\right)$

$1A = 1\left(\frac{C}{s}\right)$

$1T = 1\left(\frac{N}{A \cdot m}\right)$

Constants

$k = 9 \times 10^9 \left(\frac{N \cdot m^2}{C^2}\right)$

$\epsilon_0 = 8.85 \times 10^{-12} \left(\frac{C^2}{N \cdot m^2}\right)$

$e = 1.6 \times 10^{-19}C$

$m_e = 9.1 \times 10^{-31} \text{ kg}$

$m_p = 1.67 \times 10^{-27} \text{ kg}$

$\mu_0 = 4\pi \times 10^{-7} \left(\frac{T \cdot m}{A}\right)$

$c = 2.998 \times 10^8 \left(\frac{m}{s}\right)$

kilo = 10³, mega = 10⁶

giga = 10⁹, tera = 10¹²

milli = 10⁻³, micro = 10⁻⁶

nano = 10⁻⁹, pico = 10⁻¹²

$e^{ix} = \cos x + i \sin x$

Ch. 1

k : wave number

ω : angular frequency

$k = \frac{2\pi}{\lambda}$

$\frac{\omega}{2\pi} = f = \frac{1}{T}$

$\frac{1}{\sqrt{\mu\epsilon}} = v = \frac{\omega}{k}$

$\frac{1}{\sqrt{\mu_0\epsilon_0}} = c = \frac{\omega}{k} \quad (\text{in free space})$

$\vec{F}_E = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1q_2}{r^2}\right) \hat{r}$

$\vec{F}_B = q\vec{v} \times \vec{B}$

$\vec{E} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2}\right) \hat{r}$

$\vec{B} = \frac{\mu_0}{4\pi} \int_P \frac{Id\vec{s} \times \hat{r}}{r^2}$

$\int_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\epsilon_0}, \quad \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (= 0, \text{ in free space})$

$\int_S \vec{B} \cdot d\vec{a} = 0, \quad \nabla \cdot \vec{B} = 0$

$\int_P \vec{B} \cdot d\vec{l} = \mu_0 i_{\text{enc}} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$

$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \left(= \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}, \text{ in free space} \right)$

$\int_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0, \quad \int_S \vec{J} \cdot d\vec{a} = -\frac{dq}{dt}$

$\nabla^2 \vec{F} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{F}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{F}}{\partial t^2}$

$\vec{E} = (E_{0x}e^{i\varphi_x}\hat{x} + E_{0y}e^{i\varphi_y}\hat{y})e^{i(kz-\omega t)}$

$|\vec{E}|^2 = \vec{E} \cdot \vec{E}$

$\text{Re} \left\{ \vec{E} \right\} = E_{0x} \cos(kz - \omega t + \varphi_x) \hat{x} + E_{0y} \cos(kz - \omega t + \varphi_y) \hat{y}$

$\varepsilon = |\varphi_x - \varphi_y|$

$((\varepsilon) \bmod \pi = 0) \Rightarrow \text{linear polarization}$

$((\varepsilon) \bmod \pi = \frac{\pi}{2}) \Rightarrow \text{circular polarization}$

else $\Rightarrow$ elliptical polarization

$\frac{x^2}{E_{0x}^2} + \frac{y^2}{E_{0y}^2} - \frac{2xy}{E_{0x}E_{0y}} \cos \varepsilon = \sin^2 \varepsilon$

$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$

$\vec{B}(\vec{r}, t) = \vec{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$

$\vec{k} = k_x \hat{x} + k_y \hat{y} + k_z \hat{z} \quad (\text{dir. of wave propagation})$

$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$R_{\theta(\vec{x})} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \vec{x}$

Ch. 2

$\nabla \cdot \vec{E} = i\vec{k} \cdot \vec{E}$

$\nabla \times \vec{E} = i\vec{k} \times \vec{E}$

$\frac{\|\vec{E}\|}{\|\vec{B}\|} = \frac{\omega}{k} = c$

$$u_E=\frac{1}{2}\varepsilon_0\left\|\vec{E}\right\|^2$$

$$u_B=\frac{1}{2\mu_0}\left\|\vec{B}\right\|^2$$

$$u_E=u_B$$

$$u=u_E+u_B=\varepsilon_0\left\|\vec{E}\right\|^2$$

$$\vec{S}=(\mu_0)^{-1}(\vec{E}\times\vec{B})=\vec{E}\times\vec{H}$$

$$\frac{\partial u}{\partial t}+\nabla\cdot\vec{S}=0$$

$$\vec{S}=\left(\frac{1}{\mu_0}\right)\|\vec{E}\|\|\vec{B}\|\hat{k}=\left(\frac{1}{c\mu_0}\right)\|\vec{E}\|^2\hat{k}=cu\hat{k}$$

$$I=\langle \vec{S}\rangle=\frac{1}{2}\varepsilon_0E_0^2c$$

$$\langle u\rangle=\frac{1}{2}\varepsilon_0E_0^2$$

$$\vec{E}=\frac{1}{\mu_0\varepsilon_0}\int_0^t\big(\nabla\times\vec{B}\big)dt'$$

$$\vec{B}=-\int_0^t\big(\nabla\times\vec{E}\big)dt'$$

$$\vec{E}(\vec{r},t)=\frac{qa\sin\theta}{4\pi\varepsilon_0c^2r}\hat{\theta}$$

$$\vec{S}(\vec{r},t)=\frac{\vec{E}\times\vec{B}}{\mu_0}$$

$$|\vec{S}|=\varepsilon_0c\;|\vec{E}|^2=\frac{q^2a^2\sin^2\theta}{16\pi^2\varepsilon_0r^2c^3}$$

$$P_{\rm rad}=\frac{q^2a^2}{6\pi\varepsilon_0c^3}$$

$$P(t)=\frac{q^4\left|\vec{E}_0\right|^2\cos^2(\omega t)}{6\pi\varepsilon_0c^3}$$

$$a=q\left\|\vec{E}_0\right\|\cos(\omega t)$$

$$\langle P(t)\rangle=\frac{q^4\left\|\vec{E}_0\right\|^2}{12\pi\varepsilon_0m^2c^3}$$

$$\langle P(t)\rangle=\left(\frac{8\pi}{3}\right)\left(\frac{q^2}{4\pi\varepsilon_0mc^2}\right)^2$$

$$=\langle S\rangle_{\rm inc}=\sigma_T\langle S\rangle_{\rm inc}$$

$$\langle S\rangle_{\rm inc}=\frac{1}{2}\varepsilon_0c\;|\vec{E}_0|^2$$

$$\langle S\rangle=\langle S_0\rangle e^{-n\sigma z}$$

$$(\text{frac. scattered light})=1-e^{-n\sigma z}$$

$$P(t)=\frac{q^4\left\|\vec{E}_0\right\|^2}{6\pi\varepsilon_0m^2c^3}\left(\frac{\omega^2}{\omega^2-\omega_0^2}\right)^2\cos(\omega t)$$

$$\langle P(t)\rangle_{\rm scatt}=\sigma_T\left(\frac{\omega^2}{\omega^2-\omega_0^2}\right)^2\langle S\rangle_{\rm inc}=\sigma_{\rm ray}\;\langle S\rangle_{\rm inc}$$

$$\frac{\langle P(t)\rangle_{\rm scatt,1}}{\langle P(t)\rangle_{\rm scatt,2}}=\frac{\sigma_{\rm ray,1}}{\sigma_{\rm ray,2}}=\left(\frac{\omega_1}{\omega_2}\right)^4=\left(\frac{\lambda_2}{\lambda_1}\right)^4$$

$$(a\neq 0)\Rightarrow (\text{radiates})$$

$$(v\ll c)\Rightarrow (\text{non-relativistic})$$

Ch. 3

$$\vec{B}:=\mu_0\vec{H}+\mu_0\vec{M}=\mu\vec{H}$$

$$\vec{D}:=\varepsilon_0\vec{E}+\vec{P}=\varepsilon\vec{E}=\varepsilon_0\kappa\vec{E}$$

$$\vec{p}=qd\vec{p}$$

$$\vec{\tau}=\vec{p}\times\vec{E}$$

$$\vec{P}=n\langle\vec{p}_i\rangle=\varepsilon_0\chi_e\vec{E}$$

$$\overline{M}=n\langle\vec{m}_i\rangle=\chi_m\vec{H}$$

$$\nabla\cdot\vec{D}=0,\quad\left(\hat{n}\cdot\left(\vec{D}_1-\vec{D}_2\right)=0\Rightarrow D_1^\perp=D_2^\perp\right)$$

$$\nabla\cdot\vec{B}=0,\quad\left(\hat{n}\cdot\left(\vec{B}_1-\vec{B}_2\right)=0\Rightarrow B_1^\perp=B_2^\perp\right)$$

$$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t},\quad\left(\hat{n}\times\left(\vec{E}_1-\vec{E}_2\right)=0\Rightarrow E_1^\parallel=E_2^\parallel\right)$$

$$\nabla\times\vec{H}=\frac{\partial\vec{D}}{\partial t},\quad\left(\hat{n}\times\left(\vec{H}_1-\vec{H}_2\right)=0\Rightarrow H_1^\parallel=H_2^\parallel\right)$$

$$\frac{\left\|\vec{E}\right\|}{\left\|\vec{H}\right\|}=Z=\left(\frac{\mu}{\varepsilon}\right)^{\frac{1}{2}}$$

$$n_1\sin(\theta_i)=n_2\sin(\theta_t)$$

$$\theta_i=\theta_r$$

$$r_{\perp}=-\frac{n_1\cos(\theta_i)-n_2\cos(\theta_t)}{n_1\cos(\theta_i)+n_2\cos(\theta_t)}=-\frac{\cos(\theta_i)-\sqrt{\left(\frac{n_2}{n_1}\right)^2-\sin^2(\theta_i)}}{\cos(\theta_i)+\sqrt{\left(\frac{n_2}{n_1}\right)^2-\sin^2(\theta_i)}}$$

$$t_{\perp}=\frac{2n_1\cos(\theta_i)}{n_1\cos(\theta_i)+n_2\cos(\theta_t)}=\frac{2\cos(\theta_i)}{\cos(\theta_i)+\sqrt{\left(\frac{n_2}{n_1}\right)^2-\sin^2(\theta_i)}}$$

$$r_{\parallel}=\frac{n_2\cos(\theta_i)-n_1\cos(\theta_t)}{n_2\cos(\theta_t)+n_1\cos(\theta_i)}=\frac{\left(\frac{n_2}{n_1}\right)^2\cos(\theta_i)-\sqrt{\left(\frac{n_2}{n_1}\right)^2-\sin^2(\theta_i)}}{\left(\frac{n_2}{n_1}\right)^2\cos(\theta_i)+\sqrt{\left(\frac{n_2}{n_1}\right)^2-\sin^2(\theta_i)}}$$

$$t_{\parallel}=\frac{2n_1\cos(\theta_i)}{n_2\cos(\theta_t)+n_1\cos(\theta_i)}=\frac{2\left(\frac{n_2}{n_1}\right)\cos(\theta_i)}{\left(\frac{n_2}{n_1}\right)^2\cos(\theta_i)+\sqrt{\left(\frac{n_2}{n_1}\right)^2-\sin^2(\theta_i)}}$$

$$R=r^2,\quad T=t^2$$

$$\theta_B=\tan^{-1}\left(\frac{n_2}{n_1}\right)$$